Lecture Notes I

Course: Analytical Mechanics

Topic: Centre of mass

References: L. D. Landau and E. M. Lifshitz, *Mechanics*, 3rd ed., Section 8.

Centre of mass

Suppose that a frame $\mathscr{I}'$ moves with velocity ${}^{\mathscr{I}}v_{o'/o}$ relative to frame $\mathscr{I}$ as shown in Fig. 1.

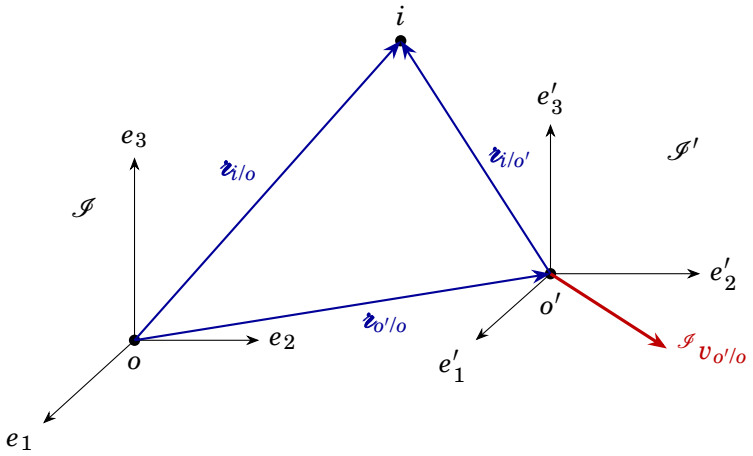


Figure 1: The frame $\mathscr{I}'$, with origin o', moves with velocity ${}^{\mathscr{I}}v_{o'/o}$ relative to the frame $\mathscr{I}$, with origin o. The position of particle i is measured from both origins, so that $\boldsymbol{r}_{i/o} = \boldsymbol{r}_{o'/o} + \boldsymbol{r}_{i/o'}$.

If we want to measure the position of particle i in both frames, we must use the equation

$$\boldsymbol{r}_{i/o} = \boldsymbol{r}_{o'/o} + \boldsymbol{r}_{i/o'}$$

so that the velocities are related by

$$\frac{{}^{\mathscr{I}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{i/o} = \frac{{}^{\mathscr{I}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{o'/o} + \frac{{}^{\mathscr{I}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{i/o'}$$

$${}^{\mathscr{I}}v_{i/o} = {}^{\mathscr{I}}v_{o'/o} + {}^{\mathscr{I}'}v_{i/o'}$$

and thus

$${}^{\mathscr{I}}p_{i/o} = {}^{\mathscr{I}'}p_{i/o'} + m_i\,{}^{\mathscr{I}}v_{o'/o}$$

where we have used $\boldsymbol{r}_{i/o'} = x_i e'_i$ and

$$\frac{{}^{\mathscr{I}'}\mathrm{d}}{\mathrm{d}t}e'_i = 0 \quad \text{(rigid)}$$

$$\frac{{}^{\mathscr{I}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{i/o'} = \dot{x}_i e'_i$$

so that for a system of particles we get (summing on i)

$${}^{\mathscr{I}}P_{T/o} = {}^{\mathscr{I}'}P_{T/o'} + \sum_{i\in S} m_i\,{}^{\mathscr{I}}v_{o'/o} \tag{1}$$

So let us look for a frame $\mathscr{I}'$ such that ${}^{\mathscr{I}'}P_{T/o'} = 0$, then equation (1) becomes (for this particular $\mathscr{I}'$)

$${}^{\mathscr{I}}P_{T/o} = \sum_{i\in S} m_i\,{}^{\mathscr{I}}v_{o'/o} = \sum_{i\in S} {}^{\mathscr{I}}p_{i/o}$$

$$ {}^{\mathscr{I}}v_{o'/o} = \frac{\sum_{i\in S} {}^{\mathscr{I}}p_{i/o}}{\sum_{i\in S} m_i} = \frac{\sum_{i\in S} m_i {}^{\mathscr{I}}v_{i/o}}{\sum_{i\in S} m_i} $$

which on integrating yields

$$ {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}R_{o'/o} = \frac{\sum_{i\in S} m_i {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{i/o}}{\sum_{i\in S} m_i} = {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\sum_{i\in S} m_i \boldsymbol{r}_{i/o}}{\sum_{i\in S} m_i}\right) $$

so that

$$ \mathbb{R}_{o'/o} = \frac{\sum_{i\in S} m_i \boldsymbol{r}_{i/o}}{\sum_{i\in S} m_i} + K $$

where K is such that $R_{o'/o}(0) = K$ is a constant of time, when looking at $R_{o'/o} = R_{o'/o}(t)$ as a function of time.

So we can make $R_{o'/o} = 0$ (that is, assuming that the origin of $\mathscr{I}'$ was the same as that of $\mathscr{I}$ at some "initial time" so $K = 0$) and

$$ R_{o'/o} = \frac{\sum_{i\in S} m_i \boldsymbol{r}_{i/o}}{\sum_{i\in S} m_i} $$

This position of this particular frame is called the centre of mass, in this course, usually denoted by G.

For a system of N-particles such that a force F acts on every particle, we can understand the same system as a system of only one particle of mass

$$ m_G = \sum_{i=1}^{N} m_i $$

located at the centre of mass on which the force acts.